

2024



REPSOL

Optimizing SAP Performance and Reliability Using AWS with Repsol

Learn how integrated energy company Repsol optimized its SAP systems using AWS.

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35% reduction

in SAP response times

40+ SAP systems

migrate



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Overview

For [Repsol](#), a global integrated energy company, a system upgrade could mean operational disruptions, lost revenue, and reputational damage. Repsol supported everything from exploration and production to retail and customer service. But with capacity limitations, agility constraints, and aging hardware, the company knew it needed an upgrade to keep up with its business.



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In 2021, Repsol signed a 7-year agreement with Amazon Web Services (AWS) to drive cloud development, innovation, and renewable energy procurement. Building on this alignment, the company migrated to [AWS for SAP](#)—the proven cloud to innovate with any SAP workload.



Opportunity | Using AWS for SAP to Modernize for Repsol

Headquartered in Madrid, Spain, Repsol has a mission to achieve net-zero emissions by 2050. To reach this goal, the company needs a robust, scalable, and agile IT infrastructure that can support its wide-ranging operations. Repsol oversees a large base of SAP systems that host its most critical environments, supporting crucial energy processes and essential business operations. However, the company faced some challenges with inherited technical debt from its existing SAP setup.

Repsol ran its SAP workloads on the cloud and on premises. The previous cloud provider had strict capacity limits, which prevented Repsol from scaling its resources as needed. On premises, the company relied on aging databases that were approaching end of life and experiencing capacity restrictions. These limitations hindered Repsol's ability to scale, innovate, and respond quickly to changing business needs.

Repsol chose AWS to host its most critical SAP workloads due to the robust performance, reliability, and security of its cloud services. To facilitate a smooth migration, the company engaged [Capgemini](#), an [AWS Partner](#) that specializes in cloud migrations, and [AWS Professional Services](#), a global team of experts that can help companies realize their desired business outcomes on AWS.

Solution | Reducing SAP Response Times by 35 Percent

Over the course of 10 months, Repsol, Capgemini, and AWS Professional Services migrated more than 40 SAP workloads to AWS. AWS Professional Services provided technical advisory services throughout the process; the team worked to verify that the migration adhered to AWS best practices and optimized the architecture design based on Repsol's business and operational requirements. Capgemini used its extensive experience in SAP migrations to facilitate a seamless implementation and guide Repsol through the complex process.

"The support provided has not only been robust but also reassuring, empowering us to feel fully backed by AWS expertise at every step of the project," says the head of SAP platforms at Repsol. "Without hesitation, we would wholeheartedly recommend AWS Professional Services to any organization seeking reliable, proactive, and expert assistance in navigating its SAP on AWS journey."

The migration involved a comprehensive assessment of Repsol's existing SAP landscape, followed by the design and implementation of a new architecture on AWS. The team used more than 140 [Amazon Elastic Compute Cloud](#) (Amazon EC2) instances, which provide secure and resizable compute capacity, to host the SAP applications. With these instances, Repsol gained the scalability and compute power that it required to handle its various workloads. Additionally, the teams attached [Amazon Elastic Block Store](#) (Amazon EBS) [gp3 volumes](#)—the latest generation of general-purpose SSD-based Amazon EBS volumes—to each Amazon EC2 instance. This setup provided reliable storage for the SAP applications, helping Repsol maintain data consistency and performance.

To help users share data across multiple instances, the teams adopted [Amazon Elastic File System](#) (Amazon EFS), a serverless, fully elastic file storage service. Amazon EFS provided a highly scalable and durable shared file system, facilitating seamless data access and collaboration across Repsol's SAP landscape. AWS Backup—a fully managed, policy-based service that simplifies data protection at scale—is used to automate backups and maintain the security and recoverability of the company's critical SAP data.

In October 2023, the 10-month migration was completed with minimal disruption to business users thanks to a series of smooth weekend cutovers. AWS Professional Services provided comprehensive training to help Repsol's delivery team efficiently manage and operate SAP workloads on AWS, facilitating a seamless transition.

As Repsol monitored its SAP systems using [Amazon CloudWatch](#), a service that gives visibility into system-wide performance, the company witnessed remarkable improvements. Some production systems experienced a 35 percent reduction in SAP response times, while nightly critical batch chains saw reduced runtimes. Developers can also provision and duplicate SAP infrastructure at a much faster pace. As a result, Repsol can deploy new SAP systems and technical upgrades faster.

Outcome | Experimenting with New Solutions on AWS

By migrating to AWS for SAP, Repsol gained the agility and capabilities necessary to stay at the forefront of innovation and adapt to evolving business needs. The improved performance and faster provisioning empowered business users to test new SAP features and experiment with new products and re



capabilities have fostered a culture of innovation within Repsol's SAP teams. Building on this success, the company plans to upgrade all SAP workloads to SAP S/4HANA.

On AWS, Repsol is well-positioned to continue its digital transformation journey. “Throughout this project, which has been one of the most challenging and complex to date, the AWS team made the difficult seem easy—a task that was far from simple,” says Repsol’s head of SAP platforms. “The planning and implementation have been truly remarkable.”

About Repsol

Repsol is a global integrated energy company headquartered in Madrid, Spain. It is committed to achieving net-zero emissions by 2050.

AWS Services Used

AWS for SAP

The proven cloud to innovate with any SAP workload.

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AWS Professional Services

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Amazon Elastic Block Store

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